

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 205 – COMPUTATIONAL METHODS IN CHEMICAL ENGINEERING
Spring 2026**

Lecture:

M/W: 11:00am – 12:15pm, EIB 124

Office Hours:

(Dr. Bilgin) M/W 12.15-1pm in person

(TA) TBD

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teaching Assistant:

Debjani Bhakta, dbhak@uic.edu

Text:

***Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB*, by V.J. Law.
ISBN: 978-1-4665-7534-9, CRC Press, 2013.**

Course Description:

Numerical and computational methods are introduced and applied to chemical engineering problems. Excel spreadsheets, VBA programming, algorithms and object-oriented concepts in chemical engineering.

Course Aims, Scope, Level:

Computer programming is playing an essential role in science and engineering curriculum. UIC Chemical engineering curriculum aims at training the students in scientific and engineering programming using different programming languages. This course introduces fundamental formulation, methodology and techniques for numerical solution of chemical engineering problems via computing. The problems are introduced in the problem-specific context, including mass and energy balance, thermodynamics, fluid flow and heat transfer. This course mainly focuses on Excel and Visual Basic for Applications (VBA). The primary reason for using Excel is its availability. Other software program packages like MATLAB require very expensive licenses for industry, therefore most likely, it will not be available to practicing engineers in industry.

Learning Objectives :

At the end of this course, a student should be able to:

- Formulate a chemical engineering problem as a mathematical model and select appropriate solution methods.
- Write and modify VBA scripts for specific computational purposes and run them.

- Use Excel functionality to tabulate thermophysical property correlations and equations of state.
- Analyze 1d (linear) and 2d (bilinear) interpolation from data tables with VBA macros.
- Calculate vector coordinates for changes of basis vectors with Excel.
- Solve ordinary differential equations (ODE) with Excel.
- Solve optimization design problems with Excel Goal Seek and Solver functionalities and VBA macros.
- Solve partial differential equations (PDE).
- Carry out error analysis of data.
- Carry out regression analysis.

Course Outline:

- Ch. 1: Roots of a single nonlinear equation
- Ch. 2: Visual Basic for applications programming
- Ch. 3: Linear algebra and systems of linear equations
- Ch. 4: Numerical differentiation and integration
- Ch. 5: Ordinary differential equations (initial value problems)
- Ch. 6: Ordinary differential equations (boundary value problems)
- Ch. 8: Partial differential equations
- Ch. 7: Regression analysis and parameter estimation
- Ch. 9: Linear programming, nonlinear programming, nonlinear equations, and nonlinear regression using Solver

Supplemental Materials:

If you need to learn more about VBA programming, you may find many tutorials on the internet. You can also consult with the following websites:

<http://excelvbatutor.com/vba tutorial.html>

<http://www.homeandlearn.org/basic mathematical operators.html>

<http://www.automateexcel.com/vba-code-examples>

Exams, Homework Assignments and Grading

There will be three midterm exams, and one final exam. All exams are required. There will be two 1 hour 15 minute-lectures per week (M-W). New material will be covered in the lecture. Students should stay current with the material and be prepared to answer questions. The teaching assistant is under no obligation to grade homework that is late, illegible, or sloppy. Scoring fewer than half of the total homework grade is grounds for failing the course. The homework is an important exercise necessary for learning the material and should be worked with care. Any evidence of copying will be addressed in accordance with the university's academic integrity policies. Assignments are due at midnight on the due date and must be submitted online through Blackboard. Late assignment (delivered before solutions are posted) will be penalized 50% on the grade of that assignment. Assignments turned in after solutions are posted will not be accepted. If there is only one late assignment throughout the semester, a late penalty will be lifted from that assignment (i.e. half of the total credit for that assignment will be returned).

Attendance

Students are expected to attend all classes, and attendance is incorporated as 15% of the final grade. Attendance credit will be assigned based on the total number of absences, regardless of whether they are excused or unexcused. Students with up to three absences will receive full (100%) attendance credit. Four absences will result in 75% attendance credit, five absences will result in 50% attendance credit, and six absences will result in 25% attendance credit. Students with more than six absences will receive a zero for attendance. Students are responsible for all material covered during any missed class sessions.

Grading Policy:

Course grades will be weighted according to the following grading formula.

Three midterms (10% each)	30%
Final exam (take-home)	10%
Homework problems (lowest dropped)	25%
Quiz (lowest dropped) (asynch weekly)	10%
Attendance/participation	15%
Sache modules	5%
In class participation	5%
Kahoot (extra credit, top three)	1%
Project (extra credit)	5%
Surveys (extra credit)	2%

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time and keep up with course materials.

Additional Information and Policies

1. EXAMS will be held during the class period. Makeup exams will not be provided except for health and weather-related emergencies.
2. Cheating on homework and exams will not be tolerated and appropriate action will be taken including (but not limited to) a failed grade or lowering of the final grade earned in the course.
3. Late Homework Policy: Homework can be turned in 1 day late without excuse for up to 50% credit.
4. Lecture Mobile Technology Policy will be given in 1st Lecture period.

Academic Integrity Policy

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy: <http://dos.uic.edu/community-standards/>

Academic Misconduct

Please see collaboration policies written above for each type of work for this class. In all cases, you cannot submit someone else's work (partial or complete) as your own. You are guilty of academic misconduct if you extend or receive any kind of unauthorized assistance. Absolutely no transfer of assignment between students is permitted (paper or electronic) under any circumstance, and you may not solicit code from other people, family, friends, or internet resources. Other examples of academic misconduct include emailing your work to another student, copying-pasting from the internet, working in a group on an assignment that states it is individual work, and allowing a tutor, TA, or another individual to write an answer or code for you. Academic misconduct is unacceptable. Minimum penalty will be a zero score on that assignment or assessment, with a letter drop to your final grade in the class. I reserve the right to exercise a change in the way grades are calculated if I deem it is necessary. More than one academic misconduct offense in this course will result in failure of the class. This means if you took this course a previous semester and were found guilty of academic misconduct, a second offense this semester will result in failure from the course. Similarly, if you have two offenses in the same semester, you will receive an F in course. Note that the Dean of Students will change your grade from W to an F and only with my approval can it remain as a W (if you were found guilty and drop the class). EVERY academic misconduct offense must be reported to the Dean of Students (if you ask me not to, you are asking me to not do my job, so please don't).

Religious Holidays

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure: <http://policies.uic.edu/educational-policy/student-academic-grievance-policy/>

Disability Accommodation

The University of Illinois at Chicago is committed to maintaining a barrier-free environment so that students with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for access to and/or participation in this course are welcome, but must be registered with the Disability Resource Center (DRC). You may contact DRC at 312-413-2183 (v) or 773-649-4535 (VP/Relay) and consult the following: <http://drc.uic.edu/guide-to-accommodations>. If you have a letter of accommodation, it is best to stop by my office hours to talk about it in person. Any exams scheduled in the DRC must overlap with the originally scheduled exam. In other words, you must start the exam before or at the same time as the originally scheduled exam. Additionally, you may not leave the DRC exam room until after the originally scheduled exam is over or later.

Mental Health and Emotional Wellness

We value your mental health and emotional wellness as part of the UIC student experience. The UIC Counseling Center offers an array of services to provide additional support throughout your time at UIC, including workshops, peer support groups, counseling, self-help tools, and initial

consultations to speak to a mental health counselor about your concerns. Please visit the Counseling Center website for more information (<https://counseling.uic.edu/>). Further, if you think emotional concerns may be impacting your academic success, please contact your faculty and academic advisers to create a plan to stay on track.

Campus Advocacy Network

If you have experienced sexual assault, domestic/dating violence, family abuse, stalking, or harassment, there is help available on campus. The Campus Advocacy Network provides free and confidential services to UIC students, faculty, and staff who have experienced or are experiencing interpersonal violence. If you would like to connect with an advocate, please contact CAN at can-appointment@uic.edu or call (312) 413-8206. To learn more, visit CAN's website at <https://can.uic.edu/>.